

Data Warehouse – 20 Tough MCQs

- Q1.** In a typical Data Warehouse architecture, the ETL process is responsible for: A) Real-time analytics
B) Data retrieval
C) Data extraction, transformation, and loading
D) Metadata generation
- Q2.** Which of the following best describes OLTP systems? A) Optimized for data analysis
B) Historical data storage
C) Real-time transaction processing
D) Complex queries over large datasets
- Q3.** OLAP cubes are used to: A) Store unstructured data
B) Normalize transactional data
C) Provide multidimensional analysis
D) Replace relational databases
- Q4.** Which modeling technique uses facts and dimensions? A) Relational modeling
B) ER modeling
C) Dimensional modeling
D) Object-oriented modeling
- Q5.** A **Data Lake** is different from a Data Warehouse in that it: A) Only stores structured data
B) Uses star schema
C) Stores raw and unprocessed data
D) Supports OLTP workloads
- Q6.** Which of the following is true about Data Warehouses? A) They support high write throughput
B) They store raw data
C) They are optimized for read-heavy queries
D) They replace OLTP systems
- Q7.** What is a Data Mart? A) A real-time data stream
B) A departmental subset of a data warehouse
C) A data normalization technique
D) A hardware component of a data warehouse
- Q8.** Star Schema is characterized by: A) Highly normalized tables
B) Fact table at center and denormalized dimension tables
C) Web of interconnected tables
D) Use of surrogate keys only
- Q9.** A Snowflake Schema is different from a Star Schema in that: A) It stores all data in a single table
B) It is highly denormalized
C) It normalizes dimension tables
D) It doesn't use fact tables
- Q10.** Which is a characteristic of an Advanced Data Warehouse? A) Only batch processing
B) No support for metadata
C) Use of AI/ML for pattern detection
D) Lack of data quality tools

- Q11.** What does a surrogate key help with? A) Creating duplicate records
B) Ensuring referential integrity
C) Maintaining natural keys
D) Reducing indexes
- Q12.** ETL in Data Warehousing usually does NOT include: A) Data cleansing
B) Data transformation
C) Data visualization
D) Data extraction
- Q13.** What distinguishes OLAP from OLTP? A) OLAP is write-optimized
B) OLAP stores raw data
C) OLAP performs complex analytical queries
D) OLTP uses star schemas
- Q14.** What kind of data does a Data Lake store? A) Only structured
B) Only unstructured
C) Only semi-structured
D) All types of data
- Q15.** Fact tables contain: A) Descriptive attributes
B) Text data
C) Numeric measures and foreign keys
D) Only primary keys
- Q16.** Which process updates a Data Warehouse with new data? A) ELT
B) OLTP
C) OLAP
D) ETL
- Q17.** Which is NOT a benefit of dimensional modeling? A) Improved query performance
B) Complex ER relationships
C) Simpler schema design
D) Easier business interpretation
- Q18.** Granularity in a data warehouse refers to: A) Number of tables
B) Data size
C) Level of detail
D) Query complexity
- Q19.** Which component holds the most detailed data? A) Data Mart
B) OLAP Cube
C) Fact Table
D) Metadata Repository
- Q20.** What is the main advantage of a Snowflake Schema? A) Better performance
B) Lower storage costs
C) More complex queries
D) More efficient space usage via normalization

Data Warehouse – Correct Answers (with your responses)

Q1. In a typical Data Warehouse architecture, the ETL process is responsible for:

✓ **Correct Answer: C)** _Data extraction, transformation, and loading_

Your Answer: C – ✓

Q2. Which of the following best describes OLTP systems?

✓ **Correct Answer: C)** _Real-time transaction processing_

Your Answer: C – ✓

Q3. OLAP cubes are used to:

✓ **Correct Answer: C)** _Provide multidimensional analysis_

Your Answer: C – ✓

Q4. Which modeling technique uses facts and dimensions?

✓ **Correct Answer: C)** _Dimensional modeling_

Your Answer: C – ✓

Q5. A Data Lake is different from a Data Warehouse in that it:

✓ **Correct Answer: C)** _Stores raw and unprocessed data_

Your Answer: C – ✓

Q6. Which of the following is true about Data Warehouses?

✓ **Correct Answer: C)** _They are optimized for read-heavy queries_

Your Answer: C – ✓

Q7. What is a Data Mart?

✓ **Correct Answer: B)** _A departmental subset of a data warehouse_

Your Answer: B – ✓

Q8. Star Schema is characterized by:

✓ **Correct Answer: B)** _Fact table at center and denormalized dimension tables_

Your Answer: B – ✓

Q9. A Snowflake Schema is different from a Star Schema in that:

✓ **Correct Answer: C)** _It normalizes dimension tables_

Your Answer: C – ✓

Q10. Which is a characteristic of an Advanced Data Warehouse?

✓ **Correct Answer: C)** _Use of AI/ML for pattern detection_

Your Answer: C – ✓

Q11. What does a surrogate key help with?

✓ **Correct Answer: B)** _Ensuring referential integrity_

Your Answer: B – ✓

Q12. ETL in Data Warehousing usually does NOT include:

✓ **Correct Answer: C)** _Data visualization_

Your Answer: C – ✓

Q13. What distinguishes OLAP from OLTP?

☒ **Correct Answer: C)** _OLAP performs complex analytical queries_

Your Answer: C – ☒

Q14. What kind of data does a Data Lake store?

☒ **Correct Answer: D)** _All types of data_

Your Answer: D – ☒

Q15. Fact tables contain:

☒ **Correct Answer: C)** _Numeric measures and foreign keys_

Your Answer: C – ☒

Q16. Which process updates a Data Warehouse with new data?

☒ **Correct Answer: D)** _ETL_

Your Answer: D – ☒

Q17. Which is NOT a benefit of dimensional modeling?

☒ **Correct Answer: B)** _Complex ER relationships_

Your Answer: B – ☒

Q18. Granularity in a data warehouse refers to:

☒ **Correct Answer: C)** _Level of detail_

Your Answer: C – ☒

Q19. Which component holds the most detailed data?

☒ **Correct Answer: C)** _Fact Table_

Your Answer: D – ☒

Q20. What is the main advantage of a Snowflake Schema?

☒ **Correct Answer: D)** _More efficient space usage via normalization_

Your Answer: D – ☒

Informatica – 20 Tough MCQs

Q1. Which of the following best describes the role of Expression Transformation?

- A) Filters rows based on condition
- B) Joins multiple sources
- C) Calculates values in a row-wise manner
- D) Aggregates data across groups

Q2. In Router Transformation, how are rows processed across different groups?

- A) All groups receive all rows
- B) Only the first matching group receives the row
- C) Rows are discarded after each group
- D) All rows go to the default group only

Q3. Which transformation is best suited to filter records based on a condition?

- A) Joiner
- B) Aggregator
- C) Filter
- D) Rank

Q4. What is the difference between Router and Filter Transformation?

- A) Router uses multiple conditions, Filter uses only one
- B) Filter drops unmatched rows, Router can route them to default
- C) Filter allows multiple outputs
- D) Router can aggregate rows

Q5. Which transformation allows you to perform aggregate calculations like SUM or AVG?

- A) Expression
- B) Aggregator
- C) Router
- D) Sequence Generator

Q6. The Joiner Transformation allows joining:

- A) Only relational tables
- B) Only heterogeneous sources
- C) Homogeneous and heterogeneous sources
- D) Only flat files

Q7. What does a Rank Transformation do?

- A) Filters NULL records
- B) Assigns a unique ID to each row
- C) Ranks rows based on a specified field
- D) Converts columns into rows

Q8. In Lookup Transformation, what is the **Connected Lookup** used for?

- A) Passes control to a prebuilt SQL script
- B) Returns multiple rows for one input
- C) Directly participates in the data flow
- D) Works only with relational sources

Q9. Which transformation is ideal for implementing Slowly Changing Dimensions Type 2?

- A) Expression
- B) Filter
- C) Update Strategy
- D) Sequence Generator

Q10. In Informatica, what does the Sequence Generator produce?

- A) Random numbers
- B) Timestamps
- C) Unique primary keys
- D) Aggregate results

Q11. Which transformation is used when data needs to be filtered *before* routing to different outputs?

- A) Aggregator
- B) Expression
- C) Filter
- D) Router

Q12. What is a key limitation of Joiner Transformation?

- A) Can't perform joins on more than two sources
- B) Can't perform outer joins

- C) Doesn't support non-equi joins
- D) Doesn't support sorting

Q13. In a connected Lookup, what happens when no match is found?

- A) Session fails
- B) NULL values are returned
- C) Row is discarded
- D) Error is raised

Q14. What does the Update Strategy Transformation decide?

- A) Sort order
- B) Transformation scope
- C) Insert, update, delete logic
- D) Rank priority

Q15. How does Informatica handle SCD Type 1?

- A) Maintains history
- B) Creates multiple rows
- C) Overwrites the existing record
- D) Adds surrogate keys

Q16. What is the default join type used by the Joiner Transformation?

- A) Inner Join
- B) Left Outer Join
- C) Right Outer Join
- D) Full Outer Join

Q17. The **Filter Transformation** evaluates conditions:

- A) Once per session
- B) Per row basis
- C) Only for input ports
- D) Only during lookup

Q18. Which transformation should be used to assign a rank to the top 3 products by revenue?

- A) Aggregator
- B) Rank
- C) Expression
- D) Lookup

Q19. Lookup Transformation cache can be:

- A) Only persistent
- B) Only dynamic
- C) Static or dynamic
- D) Only real-time

Q20. When should you use an Unconnected Lookup?

- A) When multiple rows are expected
- B) When used inside an expression or conditional
- C) When mapping output is required
- D) When updating lookup table

Informatica – Correct Answers (with your responses)

Q1. Which of the following best describes the role of Expression Transformation?

☒ **Correct Answer: C)** _Calculates values in a row-wise manner_

Your Answer: C – ☒

Q2. In Router Transformation, how are rows processed across different groups?

☒ **Correct Answer: B)** _Only the first matching group receives the row_

Your Answer: B – ☒

Q3. Which transformation is best suited to filter records based on a condition?

☒ **Correct Answer: C)** _Filter_

Your Answer: C – ☒

Q4. What is the difference between Router and Filter Transformation?

☒ **Correct Answer: B)** _Filter drops unmatched rows, Router can route them to default_

Your Answer: A – ☒

Q5. Which transformation allows you to perform aggregate calculations like SUM or AVG?

☒ **Correct Answer: B)** _Aggregator_

Your Answer: B – ☒

Q6. The Joiner Transformation allows joining:

☒ **Correct Answer: C)** _Homogeneous and heterogeneous sources_

Your Answer: B – ☒

Q7. What does a Rank Transformation do?

☒ **Correct Answer: C)** _Ranks rows based on a specified field_

Your Answer: C – ☒

Q8. In Lookup Transformation, what is the Connected Lookup used for?

☒ **Correct Answer: C)** _Directly participates in the data flow_

Your Answer: C – ☒

Q9. Which transformation is ideal for implementing Slowly Changing Dimensions Type 2?

☒ **Correct Answer: C)** _Update Strategy_

Your Answer: C – ☒

Q10. In Informatica, what does the Sequence Generator produce?

☒ **Correct Answer: C)** _Unique primary keys_

Your Answer: C – ☒

Q11. Which transformation is used when data needs to be filtered *before* routing to different outputs?

☒ **Correct Answer: D)** _Router_

Your Answer: C – ☒

Q12. What is a key limitation of Joiner Transformation?

☒ **Correct Answer: D)** _Doesn't support sorting_

Your Answer: D – ☒

Q13. In a connected Lookup, what happens when no match is found?

☒ **Correct Answer: B)** _NULL values are returned_

Your Answer: B – ☒

Q14. What does the Update Strategy Transformation decide?

☒ **Correct Answer: C)** _Insert, update, delete logic_

Your Answer: C – ☒

Q15. How does Informatica handle SCD Type 1?

☒ **Correct Answer: C)** _Overwrites the existing record_

Your Answer: C – ☒

Q16. What is the default join type used by the Joiner Transformation?

☒ **Correct Answer: A)** _Inner Join_

Your Answer: A – ☒

Q17. The Filter Transformation evaluates conditions:

☒ **Correct Answer: B)** _Per row basis_

Your Answer: C – ☒

Q18. Which transformation should be used to assign a rank to the top 3 products by revenue?

☒ **Correct Answer: B)** _Rank_

Your Answer: B – ☒

Q19. Lookup Transformation cache can be:

☒ **Correct Answer: C)** _Static or dynamic_

Your Answer: C – ☒

Q20. When should you use an Unconnected Lookup?

☒ **Correct Answer: B)** _When used inside an expression or conditional_

Your Answer: B – ☒

Tableau – 20 Tough MCQs

Q1. In Tableau, what is the difference between a *dimension* and a *measure*?

- A) Dimensions are numeric; measures are categorical
- B) Measures are filters; dimensions are views
- C) Dimensions slice the data; measures aggregate it
- D) Measures are only available in dashboards

Q2. What type of join does Tableau use by default when blending data from multiple sources?

- A) Inner Join
- B) Cross Join
- C) Left Join
- D) Full Outer Join

Q3. Which chart is best used for showing a trend over time?

- A) Pie Chart
- B) Heatmap
- C) Line Chart
- D) Treemap

Q4. What is a *LOD Expression* in Tableau used for?

- A) Filtering multiple dashboards
- B) Sorting data within filters

- C) Controlling granularity of aggregations
- D) Connecting live data sources

Q5. When should you use a Dual-Axis chart?

- A) To display text and images
- B) When comparing two measures with different scales
- C) For filtering data
- D) To link dashboards

Q6. What happens when you drag a field into the "Detail" shelf in Tableau?

- A) It filters the view
- B) It adds it as a tooltip
- C) It adds more data points to the viz
- D) It removes null values

Q7. What is the function of Tableau Extracts?

- A) To live stream data
- B) To compress and store data locally
- C) To filter real-time dashboards
- D) To delete historical data

Q8. Which of the following is *NOT* a type of calculated field in Tableau?

- A) Table Calculation
- B) Ad-Hoc Calculation
- C) Parameter Calculation
- D) LOD Expression

Q9. What does a *blue pill* represent in Tableau?

- A) A discrete field
- B) A continuous field
- C) A filter
- D) A live connection

Q10. Which of the following is **true** about context filters?

- A) They are processed last
- B) They create a temporary table for dependent filters
- C) They cannot be removed
- D) They don't affect performance

Q11. A *scatter plot* is best used when:

- A) Comparing trends over time
- B) Analyzing parts of a whole
- C) Showing correlation between two measures
- D) Displaying categories

Q12. What is a **dashboard action** in Tableau?

- A) A scheduled refresh
- B) A way to link or filter visualizations interactively
- C) A programming trigger
- D) A type of chart

Q13. What does "Show Me" in Tableau do?

- A) Automatically formats worksheets

- B) Suggests visualization types based on selected fields
- C) Imports data
- D) Filters the entire dashboard

Q14. In Tableau, what type of join should you use to include all data from both sources?

- A) Left Join
- B) Inner Join
- C) Right Join
- D) Full Outer Join

Q15. Parameters in Tableau can be used for:

- A) Sorting
- B) Dynamic filtering
- C) Calculating window averages
- D) Publishing to Tableau Server only

Q16. Which of the following allows **row-level security** in Tableau?

- A) Fixed LOD
- B) Dashboard filter
- C) User Filters
- D) Aggregation

Q17. What are Tableau Extensions used for?

- A) Creating backup files
- B) Building advanced custom visualizations using external code
- C) Converting workbooks into PDFs
- D) Publishing dashboards to Tableau Public

Q18. Which is **not** a valid function category in Tableau?

- A) Number Functions
- B) Aggregate Functions
- C) Procedural Functions
- D) String Functions

Q19. When dragging a **continuous** field to Columns shelf, Tableau creates:

- A) Discrete axis
- B) Header with labels
- C) Continuous axis
- D) Hierarchy

Q20. Which is **true** about Tableau Public?

- A) You can store private dashboards
- B) You must pay for access
- C) All workbooks are publicly viewable
- D) It doesn't support publishing

Tableau – Correct Answers (with your responses)

Q1. In Tableau, what is the difference between a *dimension* and a *measure*?

☒ **Correct Answer: C)** _Dimensions slice the data; measures aggregate it_

Your Answer: C – ☒

Q2. What type of join does Tableau use by default when blending data from multiple sources?

✓ Correct Answer: C) _Left Join_

Your Answer: C – ✓

Q3. Which chart is best used for showing a trend over time?

✓ Correct Answer: C) _Line Chart_

Your Answer: C – ✓

Q4. What is a *LOD Expression* in Tableau used for?

✓ Correct Answer: C) _Controlling granularity of aggregations_

Your Answer: C – ✓

Q5. When should you use a Dual-Axis chart?

✓ Correct Answer: B) _When comparing two measures with different scales_

Your Answer: B – ✓

Q6. What happens when you drag a field into the "Detail" shelf in Tableau?

✓ Correct Answer: C) _It adds more data points to the viz_

Your Answer: C – ✓

Q7. What is the function of Tableau Extracts?

✓ Correct Answer: B) _To compress and store data locally_

Your Answer: B – ✓

Q8. Which of the following is *NOT* a type of calculated field in Tableau?

✓ Correct Answer: C) _Parameter Calculation_

Your Answer: D – ✗

Q9. What does a *blue pill* represent in Tableau?

✓ Correct Answer: A) _A discrete field_

Your Answer: A – ✓

Q10. Which of the following is **true** about context filters?

✓ Correct Answer: B) _They create a temporary table for dependent filters_

Your Answer: B – ✓

Q11. A *scatter plot* is best used when:

✓ Correct Answer: C) _Showing correlation between two measures_

Your Answer: C – ✓

Q12. What is a **dashboard action** in Tableau?

✓ Correct Answer: B) _A way to link or filter visualizations interactively_

Your Answer: B – ✓

Q13. What does "Show Me" in Tableau do?

✓ Correct Answer: B) _Suggests visualization types based on selected fields_

Your Answer: B – ✓

Q14. In Tableau, what type of join should you use to include all data from both sources?

✓ Correct Answer: D) _Full Outer Join_

Your Answer: D – ✓

Q15. Parameters in Tableau can be used for:

✓ **Correct Answer: B)** _Dynamic filtering_

Your Answer: C – ✗

Q16. Which of the following allows **row-level security** in Tableau?

✓ **Correct Answer: C)** _User Filters_

Your Answer: C – ✓

Q17. What are Tableau Extensions used for?

✓ **Correct Answer: B)** _Building advanced custom visualizations using external code_

Your Answer: D – ✗

Q18. Which is **not** a valid function category in Tableau?

✓ **Correct Answer: C)** _Procedural Functions_

Your Answer: A – ✗

Q19. When dragging a **continuous** field to Columns shelf, Tableau creates:

✓ **Correct Answer: C)** _Continuous axis_

Your Answer: C – ✓

Q20. Which is **true** about Tableau Public?

✓ **Correct Answer: C)** _All workbooks are publicly viewable_

Your Answer: C – ✓